

F AND t TESTS FOR A GENERAL CLASS OF ESTIMATORS

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SUMMARY

To test the linear hypothesis in multiple regression we use t and F statistics. For example to test whether an individual β_i differs from zero a t-statistic is used. To test whether the entire vector β or a subvector of β is zero or equal to some pre-assigned value we use an F-statistic. In this paper it is shown that Ridge Regression (RR), Principal Components (PC) and Shrunk Estimators (SH) yield the same central t and F statistics as the Ordinary Least Squares Estimator (OLS).

1. Introduction

Consider the linear model

(1.1) $Y = \beta_0 1 + X\beta + \epsilon$ where Y is an $n \times 1$ vector of response variables; 1 is an $n \times 1$ vector of ones; $X = (X_1, X_2, \dots, X_p)$ is a full-column rank matrix, i.e. $r(X) = p$, of non-stochastic regressor variables, which are standardized so that $X_j'1 = 0$ and $X_j'X_j = 1$ for $j = 1, 2, \dots, p$; β_0 is an unknown scalar constant, while β is a $p \times 1$ vector of unknown regression coefficients; ϵ is an $n \times 1$ vector of unobservable random error variables with $\epsilon \sim N(0, \sigma^2 I)$.

Denote the latent roots of $X'X$ by $0 < \ell_1 \leq \ell_2 \leq \dots \leq \ell_p$ and the corresponding orthonormal latent vectors by $V_1, V_2, \dots, V_p$. Small latent roots and the corresponding latent vectors identify multicollinearities. This is because the $\ell_j = V_j'X'XV_j$ for $j = 1, 2, \dots, p$. It therefore follows that if r out of the p -latent roots are suitably small the above relation becomes

$$0 \approx (XV_j)'(XV_j) \text{ and so } XV_j \approx 0 \text{ for } j = 1, 2, \dots, r.$$

The problem of multicollinearity can be overcome by using biased estimation procedures such as Ridge, Principal Components, etc.

Consider the general form of the OLS, PC, SH and RR estimators which is given by (see for example Gunst and Mason (1977)).

$$(1.2) \quad \tilde{\beta} = \sum_{j=1}^p a_j C_j V_j$$

where $C_j = V_j' X' Y$ is the same for all estimators and a_j depends on the particular estimator. More specifically if (Gunst and Mason (1977))

$$(1.3) \text{ (i)} \quad \begin{aligned} a_j &= \ell_j^{-1} \text{ then } \tilde{\beta} \text{ is the OLS estimator} \\ \hat{\beta}' &= (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p); \end{aligned}$$

$$\text{(ii)} \quad a_j = (\ell_j + k)^{-1} \text{ with } k > 0, \text{ then } \tilde{\beta} \text{ is the RR estimator;}$$

$$\text{(iii)} \quad a_j = d^* \ell_j^{-1} \text{ where } 0 \leq d^* < 1, \text{ then } \tilde{\beta} \text{ is the deterministic SH estimator (although the choice of } d^* \text{ depends on the data, in this paper it is treated as a constant);}$$

$$\text{(iv)} \quad a_j = 0 \text{ for } j = 1, 2, \dots, r \text{ and } a_j = \ell_j^{-1} \text{ for } j = r+1, \dots, p \text{ then } \tilde{\beta} \text{ is the PC estimator.}$$

It is shown below that the t and F statistics for testing the general linear hypothesis using the estimator in (1.2) are the same for any choice of a_j ((1.3)(i) – (1.3)(iv)). Moreover we state some results concerning the sign of the estimator $\tilde{\beta}$ given in (1.2).

It is important to stress that the theory which is presented here is classical rather than Bayesian. In what follows we make use of the Constrained Least Squares estimator (CLS). The CLS estimator, when β satisfies the relation

$$(1.4) \quad H\beta = h$$

is given by (Goldberger (1964))

$$(1.5) \quad \hat{\beta}_c = \hat{\beta} - (X'X)^{-1} H' (H(X'X)^{-1} H')^{-1} (H\hat{\beta} - h)$$

Note that H is a known $r \times p$ matrix of rank r , i.e. $r(H) = r$; $1 \leq r \leq p$, and h is a known $r \times 1$ vector.

If $\hat{\beta}_i = 0$, then the H matrix in (1.4) and (1.5) is a row vector with one in the i^{th} column and zeroes elsewhere. We can then write the estimator $\hat{\beta}_c$ in (1.5) as

$$(1.6) \quad \hat{\beta}_{ch} = \hat{\beta} - \Delta_i^{-1} \hat{\beta}_i / (X'X)_{ii}^{-1}$$

where Δ_i^{-1} is the i^{th} column (row) of $(X'X)^{-1}$ while $(X'X)_{ii}^{-1}$ is the i^{th} diagonal ele-

ment of $(X'X)^{-1}$.

2. The t-statistic for the general estimator

Suppose that we are interested in testing the hypothesis

$$(2.1) \quad H_0 : \beta_i = 0 \text{ against } H_1 : \beta_i \neq 0$$

A t-statistic for testing (2.1) can be formed as follows. The least squares t-statistic for testing (2.1) is known to be

$$(2.2) \quad t_i = \frac{(\hat{\beta}_i - 0)}{\{S^2(X'X)^{-1}_{ii}\}^{1/2}}$$

where $(X'X)^{-1}_{ii}$ is that in (1.6) and S^2 is the least squares estimate for σ^2 .

Note that the numerator of t_i is the i^{th} element of $\hat{\beta}$ minus zero, which is the expected value of $\hat{\beta}_i$ under H_0 . The denominator is the square root of an unbiased estimator of the variance of the numerator. If we assume normality, under H_0 , t_i of (2.2) has the central t-distribution with $n-p-1$ degrees of freedom.

A t-statistic for testing the hypothesis in (2.1) using $\tilde{\beta}_i$, which is the i^{th} component of $\tilde{\beta}$ in (1.2), for the cases (1.3)(ii) to (1.3)(iv), can be formed in direct analogy with (2.2), but it must be remembered that $\tilde{\beta}_i$ is biased (in this respect see also Obenchain (1977)).

For the i^{th} element of $\tilde{\beta}$ we have

$$\begin{aligned} (2.3) \quad \tilde{\beta}_i &= \sum_{j=1}^p a_j V_{ij} C_j \\ &= \sum_{j=1}^p a_j V_{ij} (V_j' X' Y) \\ &= \sum_{j=1}^p a_j V_{ij} (V_j' X' X \hat{\beta}) \end{aligned}$$

where V_{ij} is the i^{th} element of the j^{th} latent vector. If $\hat{\epsilon}_i$ denotes the linear unbiased, minimum variance estimator of $E(\tilde{\beta}_i)$, when $\beta_i = 0$ holds, then $\hat{\epsilon}_i$ is given by

$$\begin{aligned} (2.4) \quad \hat{\epsilon}_i &= \sum_{j=1}^p a_j V_{ij} (V_j' X' X \hat{\beta}_{ch}) \\ &= \sum_{j=1}^p a_j V_{ij} (V_j' X X [\hat{\beta} - \Delta^{-1}_i \hat{\beta}_i / (X'X)^{-1}_{ii}]) \end{aligned}$$

with $\hat{\beta}_{ch}$ given in (1.6).

The t-statistic for testing the hypothesis (2.1) is analogous to that of (2.2). Using $\tilde{\beta}_i$ of (2.3) this statistic is given by

$$(2.5) \quad t_i^* = \frac{\tilde{\beta}_i - \hat{e}_i}{\{\text{Var}(\tilde{\beta}_i - \hat{e}_i)\}^{1/2}}$$

For the numerator of (2.5), using (2.3) and (2.4), we have

$$\begin{aligned} (2.6) \quad \tilde{\beta}_i - \hat{e}_i &= \{\sum_{j=1}^P a_j V_{ij} (V_j' X' X \hat{\beta})\} - \{\sum_{j=1}^P a_j V_{ij} (V_j' X' X \\ &\quad [\hat{\beta} - \Delta_i^{-1} \hat{\beta}_i / (X' X)_{ii}^{-1}])\} \\ &= \sum_{j=1}^P a_j V_{ij} \{(V_j' (X' X) \hat{\beta}) - (V_j' (X' X) [\hat{\beta} - \Delta_i^{-1} \hat{\beta}_i / (X' X)_{ii}^{-1}])\} \\ &= \sum_{j=1}^P a_j V_{ij} V_j' (X' X) \{\Delta_i^{-1} \hat{\beta}_i / (X' X)_{ii}^{-1}\} \\ &= \sum_{j=1}^P a_j V_{ij} ((X' X) V_j)' \{\Delta_i^{-1} \hat{\beta}_i / (X' X)_{ii}^{-1}\} \\ &= \sum_{j=1}^P a_j V_{ij} (Q_j V_j)' \{\Delta_i^{-1} \hat{\beta}_i / (X' X)_{ii}^{-1}\} \\ &= \sum_{j=1}^P a_j V_{ij} Q_j ((\Delta_i^{-1})' V_j)' \hat{\beta}_i / (X' X)_{ii}^{-1} \\ &= \sum_{j=1}^P a_j V_{ij} Q_j (Q_j^{-1} V_{ij})' \hat{\beta}_i / (X' X)_{ii}^{-1} \\ &= \sum_{j=1}^P a_j V_{ij}^2 \hat{\beta}_i / (X' X)_{ii}^{-1} \\ &= d_i^2 \hat{\beta}_i / (X' X)_{ii}^{-1} \end{aligned}$$

where

$$d_i^2 = \sum_{j=1}^P a_j V_{ij}^2$$

In addition we have

$$\begin{aligned} (2.7) \quad \text{Var}(\tilde{\beta}_i - \hat{e}_i) &= \text{Var}(d_i^2 \hat{\beta}_i / (X' X)_{ii}^{-1}) \\ &= (d_i^2 / (X' X)_{ii}^{-1})^2 \sigma^2 (X' X)_{ii}^{-1} \\ &= \sigma^2 d_i^4 / (X' X)_{ii}^{-1} \end{aligned}$$

If the σ^2 in (2.7) is estimated by its unbiased estimator S^2 and we substitute together

with (2.6) into (2.5) we obtain

$$\begin{aligned}
 (2.8) \quad t_1^* &= \frac{(d_1^2 \hat{\beta}_1 / (X'X)_{11}^{-1})}{(S^2 d_1^4 (X'X)_{11}^{-1})^{1/2}} \\
 &= \hat{\beta}_1 / (S^2 (X'X)_{11}^{-1})^{1/2} = t_1
 \end{aligned}$$

The d_1^2 is always positive when at least one of the $a_j > 0$ for $j = 1, 2, \dots, p$.

The results of Leamer (1975) generalise to the following.

1. The estimate of β_j using the general estimator (1.2) and the CLS estimate of β_j , as given by (1.6), with $\beta_i = 0$ for $i \neq j$, have the same sign if the t-statistic of β_j is less in absolute value than the t-statistic of β_j .
2. Let $\tilde{\beta}$ be given as in (1.2) and the CLS estimate $\hat{\beta}_c$ be given by (1.5). Then the estimate of β_j given by $\tilde{\beta}_j$ and the constrained least square estimate of β_j , $\hat{\beta}_{cj}$, with the constraint $H\beta = h$, satisfy the inequality

$$|\tilde{\beta}_j - \hat{\beta}_{cj}| \leq (S^2 (X'X)_{jj}^{-1})^{1/2} r^{1/2} u_1^{1/2}$$

where r is the rank of H and u_1 is the statistic for testing the constraint $H\beta = h$ and is defined by (3.4) in Section 3 of this paper.

3. The F-Test for the General Estimator

Suppose we want to test

$$(3.1) \quad H_0 : H\beta = h \text{ against } H_1 : H\beta \neq h$$

with H, h as in (1.5). Clearly we have $E(H\tilde{\beta}) \neq h$ when $\tilde{\beta}$ is not the OLS estimator of β .

Let $\hat{e} = H \sum_{j=1}^p a_j V_j V_j' (X'X) \hat{\beta}_c$ be the linear unbiased minimum variance estimator of $E(H\tilde{\beta})$, when H_0 is true. Note that if $E(H\tilde{\beta})$ is known then $\text{Var}(\hat{e}) = 0$. Consider now

$$\begin{aligned}
 (3.2) \quad u^* &= H\tilde{\beta} - \hat{e} \\
 &= H \sum_{j=1}^p a_j V_j V_j' (X'X) \tilde{\beta} - H \sum_{j=1}^p a_j V_j V_j' (X'X) \hat{\beta}_c
 \end{aligned}$$

$$\begin{aligned}
&= H \sum_{j=1}^p a_j V_j V_j' (X'X) [\hat{\beta} - \hat{\beta}_c] \\
&= H \sum_{j=1}^p a_j V_j V_j' (X'X) [\hat{\beta} - \{\hat{\beta} - (X'X)^{-1} H' (H(X'X)^{-1} H')^{-1} (H\hat{\beta} - h)\}] \\
&= H \sum_{j=1}^p a_j V_j V_j' H' (H(X'X)^{-1} H')^{-1} (H\hat{\beta} - h) \\
&= T u
\end{aligned}$$

where

$$T = H \sum_{j=1}^p a_j V_j V_j' H' (H(X'X)^{-1} H')^{-1} \text{ and } u = H\hat{\beta} - h.$$

Under H_0 we have

$$\begin{aligned}
(3.3) \quad (i) \quad E(u^*) &= 0 \\
(ii) \quad \text{Var}(u^*) &= T \text{Var}(u) T' \\
&= T \sigma^2 (H(X'X)^{-1} H') T' \\
&= T Q T' = Q^*
\end{aligned}$$

where

$$Q = \sigma^2 H(X'X)^{-1} H'.$$

If we substitute σ^2 , by its estimate S^2 , in (3.3)(ii) then the F-statistic for testing (3.1) will be

$$\begin{aligned}
(3.4) \quad F^* &= \frac{u^{*'} Q^{*-1} u^*}{r} \\
&= \frac{u' T' (T')^{-1} Q^{-1} (T)^{-1} T u}{r} \\
&= \frac{u' Q^{-1} u}{r} \\
&= \frac{(H\hat{\beta} - h)' (H(X'X)^{-1} H')^{-1} (H\hat{\beta} - h)}{r S^2} \\
&= \frac{(\hat{\beta} - \hat{\beta}_c)' (X'X) (\hat{\beta} - \hat{\beta}_c)}{r S^2} = F
\end{aligned}$$

In view of the above the following comments are made:

(i) The last relation represents the F-statistic for testing (3.1) when we make

use of the least squares estimates for β .

- (ii) The $F^* = F$ in (3.4) is distributed as F with $r, n-p-1$ degrees of freedom.
- (iii) A sufficient condition for the matrix T to be non-singular is that $a_j > 0$ for $j = 1, 2, \dots, p$. In the case where $\tilde{\beta}$ is the PC estimator the inverse of T exists provided that $p - r \geq r$, i.e. $p > 2r$.

Using the above results it is easy to see that the confidence interval centred at $\hat{\beta}_i$ is the same for the OLS, PC, RR, SH estimators. It is important to point out that confidence intervals which are centred at biased estimators have greater maximum diameter than the least squares interval of the same confidence. (Refer Stein (1962)).

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